INCH-POUND

MIL-C-24643/12C
14 March 1994
SUPERSEDING
MIL-C-24643/12B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSTTOP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor - AWG size 20, class K stranding, uncoated (see note 1).
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
* Third - Ethylene propylene rubber or cross-linked polyethylene insulation. Minimum average wall thickness 0.020 inch. Telephone identification code shall be applied by method 3.
Fourth - Two conductors shall be cabled together to form pairs, with a lay suitable to meet test requirements specified herein.
Fifth - The specified number of pairs (see table I) shall be cabled together in one or more layers, all in left hand direction, with the minimum amount of fillers necessary, to obtain a firm well-rounded assembly. Cabling sequence to be consecutive, starting with no. 1, from center outward. The length of lay shall be determined to permit a completed cable to meet the twisting and bending endurance tests. A single pair shall not be used as center for the cable core.
Sixth - Binder tape applied helically with overlap.
Seventh- Cross-linked polyolefin jacket. (see table I for wall thickness).

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Note: 1. At the manufacturer's option, conductors may consist of eight strands, each having a nominal diameter of 0.010 inch, around one strand having a nominal diameter of 0.0142 inch.

TABLE I. Details

Military part no. M24643/12	Type and size	Number of pairs	Cable jacket thickness min avg (inch)	Overall diameter maximum (inch)
-01UN	LSTTOP-3	7	0.060	0.480
-02UN	LSTTOP-5	10	.060	.590
-03UN	LSTTOP-10	14	.060	.700
-04UN	LSTTOP-15	19	.060	.830

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum..	11.53
Voltage withstand - volts, root mean square, minimum conductor to conductor.....	2000
Insulation resistance - conductor to conductor, megohms/1000 feet, minimum.....	100
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
* Group B:	
Cross-linked proof test (percent, maximum)	
Insulation.....	50
Jacket (When tested at 200°C).....	50
Tear - pounds per inch thickness, minimum (ASTM D 470).....	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum.	700
Elongation - percent, minimum.....	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum.	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.	1300
Elongation - percent, minimum.....	160

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EXAMINATION AND TESTS (continued):	Requirements
Twisting and bending endurance - (see note 2)	
At 75°C, cycles, minimum	6000
At minus 30 ±2°C, cycles, minimum	2000

Note: 2. Roller diameter for bending endurance test shall be as follows:

Cable type and size	Roller diameter (inches, maximum)
LSTTOP-3	1.4
LSTTOP-5	1.8
LSTTOP-10	2.1
LSTTOP-15	3.0

Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Shrinkage	No failure
* Permanence of printing (jacket) - cycles, minimum	125
Heat distortion - percent of unaged, maximum	30

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EXAMINATION AND TESTS (continued): Requirements

Group D:

*	Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable) (125 ±5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
3, 5 and 10	2500
15	1500

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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